Singular Gram Certificates and Relative Errors in the Laughlin Fock-Space Inequality

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Abstract

This confidential report audits the pinned manuscript *A Fock-space inequality and the Laughlin spectral gap* at the level of its operator spaces, normal-ordering reductions, spin multiplicities, rational certificates, and finite-sphere transfer. It does not independently verify the claimed uniform Laughlin gap. We give a rational rank-reduced replacement for the singular four-body Gram test, an explicit finite-flux pair-energy perturbation budget, and an anisotropic allowance principle that retains particle-number uniformity. A separate local extension uses spherical singlet symmetry to compute exactly the particle-addition overlap projector and sharpen the source's charge-versus-neutral comparison without using its uniform-gap theorem. The author's bounded exact arithmetic reproduces all thirteen retained three-body margins, all twenty-three limiting Gram ranks, and the first nine four-body tests; an independent sidecar reproduces all twenty-three four-body tests and supplies a finite-flux support refinement reducing the allowance count from 1222 to 1123. Finite-flux numerical thresholds and the global source conclusion remain outside the executed verification. The private Ram correspondences concern physical metrics, compatible analysis and synthesis, and residual-aware certificates; they are not replacements for fermionic algebra or spin representation theory. Contribution assessment: C3/5, exact quotient certificates and explicit relative-error accounting together with a proved spherical overlap refinement constitute a notable local contribution while finite-flux thresholds and the source headline theorem remain unverified.

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1 Scope, pin, and mathematical contract

The source [S] is OpenAI's September 24, 2026 manuscript with catalogue ID 546 and directory

> `preprints/A-Fock-space-inequality-and-the-Laughlin-spectral-gap-September-24-2026`.

The catalogue-declared pin is

> `adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

No remote byte-for-byte checkout authentication was performed; the recorded local SHA-256 hashes identify the actual inputs. The local study root is `C:/Users/mural/OneDrive/Documents/OpenAI-722-math`; source files are under `_sources/openai-math/` followed by that directory. The row's L5 applicability is not the report's realized contribution score. The contribution policy used is `contribution-policy.json`. All substantive source sections and the mathematical appendix were read, including the row data and bibliography. Exact byte hashes and the read-scope record accompany this report; reading a proof is distinguished throughout from independently verifying every dependency of it.

1.1 Spaces, normalization, and boundaries

For each integer $Q \geq 1$, U_Q is the complex spin-$Q/2$ irreducible unitary representation of $SU(2)$, of dimension $Q+1$. It can be realized as homogeneous spinor polynomials of degree Q, with invariant Hermitian inner product and normalized lowering basis $e_0, \ldots, e_Q$. The physical geometry is the round sphere: it has no spatial boundary, and no hard orbital cutoff is substituted for its full pair interaction. The finite lowest-Landau-level space at flux Q is the model itself. Use the convention that increasing wedges of distinct orthonormal one-particle vectors have norm one. Then

$$\mathcal{F}_Q = \bigoplus_{n=0}^{Q+1} \bigwedge^n U_Q, \qquad H_Q = \bigoplus_{n=0}^{Q+1} H_{n,Q}, \qquad H_{n,Q} = \sum_{1 \leq i < j \leq n} P_{ij}^{(1)}.$$

Here $P_{ij}^{(1)}$ is the orthogonal pair-spin-$(Q-1)$ projector in tensor positions i, j, restricted after summation to the antisymmetric space. Each unordered pair has coefficient *one*; $H_{0,Q} = H_{1,Q} = 0$. All spherical operators are everywhere-defined bounded operators on finite-dimensional Hilbert spaces. In particular, a form inequality means an inequality for every vector, not just basis vectors, selected particle numbers, or sampled states.

The pair-annihilation convention is $B(v) = (v \wedge \cdot)^*$, conjugate linear in v, and $c_j = B(e_j)$. Thus $\{c_i, c_j^*\} = \delta_{ij} I$ and $B(v \wedge w) = B(w)B(v)$. For the spin-$(Q-1)$ pair subspace $V_Q \subset \bigwedge^2 U_Q$ and orthonormal basis $v_0, \ldots, v_{2Q-2}$,

$$H_Q = \sum_{p=0}^{2Q-2} B_p^* B_p, \qquad B_p = B(v_p), \qquad d_a = 2Q - 1. \tag{1}$$

Haar measure on $SU(2)$ has mass one. No unspecified measure volume or particle-dependent energy scale is absorbed in a constant.

1.2 Precisely what the source claims

With

$$\gamma_* = \frac{4616733319001}{10^{14}}, \tag{2}$$

source Theorem `thm:fock` states

$$\forall \gamma \in (0, \gamma_*) \quad \exists Q_\gamma \in \mathbb{N} \quad \forall Q \in \mathbb{N}, Q \geq Q_\gamma : \quad H_Q^2 \succeq \gamma H_Q.$$

The threshold is independent of particle number, not explicit, and does not assert the endpoint on a finite sphere. A positive eigenvalue is then at least γ. In a sector with no zero mode this is a lower bound on its lowest energy, not on its first excitation above that positive lowest energy. At $Q = 3(N-1)$ the known unique kernel is

$$\mathbb{C}\Psi_N, \qquad \Psi_N = \prod_{i<j} (u_i v_j - u_j v_i)^3,$$

and the source concludes $H_{N,Q} \succeq (1/25)(I - P_N)$ for sufficiently large N, where P_N is the orthogonal projector onto its normalized Laughlin line. This report does not promote that source conclusion to an independently established result.

For its planar corollary, the source uses the Bargmann space

$$\mathcal{B} = \left\{ f \text{ entire} : \ \frac{1}{\pi} \int_{\mathbb{C}} |f(z)|^2 e^{-|z|^2} \, d^2 z < \infty \right\}, \quad e_j(z) = z^j / \sqrt{j!}.$$

For fixed N, H_N^{pl} is the sum of the unrestricted relative-degree-one pair projectors on $\bigwedge^N \mathcal{B}$. It is bounded on that entire space by $\binom{N}{2} I$. Gaussian square-integrability replaces any artificial boundary condition at infinity. The source endpoint $(H_N^{\mathrm{pl}})^2 \succeq \gamma_* H_N^{\mathrm{pl}}$ has quantifiers over *every* $N \geq 2$ and total degree L, and is conditional here on the spherical source theorem.

2 Stepwise source dependency audit

We use source labels to make the audit independent of page reflow.

2.1 Exterior algebra and the normal-ordered square

Source `build/sections/02-operators.tex`, `lem:square`, uses the positive lift

$$\mathcal{L}_k(|v\rangle\langle w|) = B(v)^* B(w).$$

Its normalization on the k-particle sector is the original matrix. On N particles, contraction in two specified tensor slots corresponds to $B(v)/\sqrt{\binom{N}{2}}$, explaining (1). Define wedge maps $W_3 : V_Q \otimes U_Q \to \bigwedge^3 U_Q$ and $W_4 : V_Q \otimes V_Q \to \bigwedge^4 U_Q$. The source identity is

$$H^2 = H + \mathcal{L}_3(W_3(W_3^* W_3 - I)W_3^*) + \mathcal{L}_4(W_4 W_4^*). \tag{3}$$

Two middle CAR contractions give the two-body term; zero contractions give an *ordered* pair sum and hence $W_4 W_4^*$ without an extra factor of two. On three particles $H = W_3 W_3^*$, determining the three-body term after subtracting H. Successive particle-sector restrictions determine normal-ordered coefficients. The logic is source-specific and sound at this algebraic level; a finite spectrum at one N would not substitute for this identity on all $\mathcal{F}_Q$.

2.2 Pair coefficients, phases, and the three-body spectrum

Source `03-spin.tex` fixes $v_0 = -e_0 \wedge e_1$ and the lowering coefficient $\sqrt{(p+1)(2j-p)}$ in a spin-j representation. With $(s)_p = s(s-1)\cdots(s-p+1)$, its formulas are

$$v_p(i,j) = (i-j)\sqrt{\frac{(Q)_i (Q)_j p!}{Q(2Q-2)_p i! j!}}, \qquad i<j, \quad i+j = p+1, \tag{4}$$

$$v_p^*(i,j) = (i-j)\sqrt{\frac{p!}{2^p i! j!}}, \qquad i<j, \quad i+j = p+1. \tag{5}$$

All other coefficients vanish; the star in v_p^* denotes a limit, not an adjoint. Binomial differentiation verifies normalization. In the multiplicity-free decomposition of $V_Q \otimes U_Q$, let $R_{3,z}$ project onto spin $3Q/2 - 1 - z$, $0 \leq z \leq Q$. Then

$$W_3^* W_3 - I = \sum_z q_z(Q) R_{3,z}, \quad q_z(Q) = (-1)^z \frac{(Q)_z}{(2Q-2)_z}\left(3z - 1 - \frac{z(z+1)}{Q}\right). \tag{6}$$

The sectors $z = 0, 1, 3$ are wedge-null when allowed. The source derives this by compressing $I - P_{13} - P_{23}$ and evaluating a highest-vector coefficient, not by numerical eigenvalue fitting.

For fixed deficits, coupled coefficients converge to the normalized monomial coefficients of

$$\frac{(uX - vY)^z (vX + uY)^{T-z}}{\sqrt{z!(T-z)!}}, \qquad v = \sqrt{1-u^2}. \tag{7}$$

Here u^2 is the limiting fraction of the second spin in the sum. The three-body case uses $u = 1/\sqrt{3}$, the nested four-body cases $u = 1/\sqrt{2}$. Normalized lowering must match the equivalent copies: phase inconsistency in those copies would alter off-diagonal matrices.

2.3 Seven squares and multiplicity-sensitive averaging

In source `04-comparison.tex`, seven rows define

$$F_\rho = \lambda_\rho B_{t_\rho} + \sum_{p,j} \alpha^\rho_{pj} c^*_{p+j-t_\rho} c_j B_p, \qquad K = \sum_{\rho=1}^{7} F^*_\rho F_\rho. \tag{8}$$

The ranges are $0 \le p \le 7$, $0 \le j \le 8$, $0 \le p + j - t_\rho \le 8$. All terms remove two particles and orbital weight $t_\rho + 1$. CAR normal ordering gives a two-body cost ηH, three-body cross and contraction terms, and a negative four-body term. With $i = p + j - t$, $k = q + l - t$ the latter is

$$-\sum_{p,j;q,l} \alpha_{pj}\alpha_{ql} (c_k c_j B_p)^* (c_i c_l B_q).$$

The positive rotational average $d_a \int K(g)\, dg = A_2 + A_3 + A_4$ has $A_2 = \eta H$, $\eta = \sum_\rho \lambda_\rho^2$. Schur averaging on a spin factor is I_d/d times its partial trace; it retains off-diagonal entries between equal-spin copies. Replacing it by entrywise diagonalization in the copy index is not legitimate.

Three-body deficits retained are $z \le 15$, with limits e_z and dimension $d_{3,z} = 3Q - 1 - 2z$. The strict tests are

$$e_z < \frac{3}{2}(3z - 1)(-1/2)^z, \qquad z \in \{2, 4, 5, \ldots, 15\}. \tag{9}$$

They imply the finite-flux three-body comparison for all sufficiently large Q, because there are finitely many retained blocks and the other three blocks are exactly wedge-null. Section 8 replays every limiting inequality, not its finite-flux threshold.

Four-body deficit $D \le 23$ has copy labels $\mathcal{R}_D = \{1, 3, \ldots\} \cap [1, D]$ and spin dimension $d_D = 4Q - 1 - 2D$. With $\widehat{W} : V_Q \otimes \bigwedge^2 U_Q \to \bigwedge^4 U_Q$, write

$$w^Q_{Dr} = \widehat{W}|D, D; r\rangle_Q, \quad G_{D,rs} = \langle w^*_{Dr}, w^*_{Ds}\rangle, \quad \chi_{rs} = \mathbf{1}_{r=s=1}.$$

The source copy matrices E^Q_D converge to E_D. The target is the $r = 1$ column with coefficient $d_D/d_a \to 2$. For even D that column wedges to zero, since the two pair factors are antisymmetric under exchange but exterior multiplication of degree-two forms is symmetric. The source does not assume full rank. Its limiting tests are

$$G_D(2\chi - E_D + \varepsilon I)G_D \succeq 0, \qquad 1 \le D \le 23, \quad \varepsilon = 3/10^6. \tag{10}$$

Deficits above 23 in the target four-body term are positive and may be discarded; this does not permit discarding the negative three-body tail.

2.4 Relative transfer, tail, kernel, and plane

Source `05-transfer.tex`, `lem:transfer`, works first on $\mathcal{F}_{\mathrm{loc}} = \bigwedge^* \mathbb{C}^{25}$, modes $0, \ldots, 24$. It obtains each retained four-body error as $-\rho_Q h_Q$, $h_Q = \sum_{p=0}^{23} B^*_p B_p$, with $\rho_Q \to 0$. Ordering local modes before spectators makes the local annihilators local operators tensored with the identity. The parity factor belongs to later spectator annihilators, which do not occur. Thus the estimate extends to all particle numbers without a dimension factor. Averaging gives $d_a \int h_Q(g)\, dg = 24 H_Q$.

Source Proposition `prop:A4` pays for the allowance by Bessel's inequality before wedging. The orthonormal highest-copy coefficient matrix has $SS^{*} = I$, though the w_{Dr}^{Q} need not be orthogonal. The number of triples $p + j + k = D$, $j < k$, is

$$n_D = \sum_{n=1}^{D} \lceil n/2 \rceil = \left\lfloor \frac{(D+1)^2}{4} \right\rfloor, \quad \sum_{D=1}^{23} n_D = 1222.$$

Hence the allowance costs at most $1222\varepsilon H_Q$, independently of N.

Source `06-gap.tex` bounds the negative tail by $-\delta_Q H$,

$$\delta_Q = \sum_{\substack{17 \le z \le Q \\ z \text{ odd}}} \frac{3Q - 1 - 2z}{2Q - 1} |q_z(Q)|(z+1) \longrightarrow \frac{61}{4096}. \tag{11}$$

For $Q \ge 4$ a summable majorant is $6z(z+1)(2/3)^z$. Indeed only odd z are negative in this range and the bracket in (6) is at least $2z - 2$. Substituting $z = 17 + 2m$ gives the limiting numerator $900 + 208m + 12m^2$ and the geometric sums $4/3, 4/9, 20/27$, proving the limit. Combining the comparisons yields

$$H_Q^2 \succeq d_a \int K(g)\, dg + (1 - \eta - \delta_Q - 1222\varepsilon - o(1))\, H_Q, \tag{12}$$

where $\eta = 93527408868499/10^{14}$ and the limiting coefficient is (2). This implication depends on *all* tests (10), not just the nine blocks replayed by the author. The independent full finite-certificate replay is described separately in Section 8.3.

The source's kernel proof uses the irreducible pair brackets $b_{ij} = u_i v_j - u_j v_i$. An antisymmetric pair without relative degree one lies in odd relative degrees at least three and is divisible by b_{ij}^3. Distinct brackets are nonassociate irreducibles in the polynomial unique-factorization domain. Their product divides a zero mode; at $Q = 3(N-1)$ it exhausts each particle multidegree and leaves only a constant. Conversely the product has no pair-spin $Q - 1$ component. This elementary kernel result, not the source gap theorem, is used independently in Section 7.

Source `07-plane.tex` fixes N, L before taking $Q \to \infty$. For $Q \ge \max(L, N-1, 2)$ the lowering-deficit-L sector is the same finite orthonormal Slater basis as the planar degree-L sector. Both are invariant and their pair coefficients converge in norm. For each fixed $\gamma < \gamma_*$, the spherical inequality passes to that block; only then is $\gamma \uparrow \gamma_*$ taken. The bounded direct sum over L completes the fixed-N planar assertion. This is not an appeal to strong resolvent convergence on growing arbitrary compressions. Neither norm convergence uniformly in L nor an explicit $Q(N, L, \gamma)$ is supplied or needed in that conditional argument.

3 Private Ram correspondences and their limits

The following are actual local passages read for this report, not routing-map substitutes. Paths are confidential and were never submitted to public web searches. Only the listed passages, not their entire larger books, are claimed as corpus reading.

3.1 Physical Gram metrics

`C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex`

The entire chapter fragment was read, in particular `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, `v21c01:residual-decomposition`, `v21c01:residual-identity`, and `v21c01:gram-equivalence`. It declares physical synthesis J, metric $G = J^*J$, and the omitted residual $(I - P)AJc$. Its initial inverse-Gram formulas explicitly assume injective J. The present wedge synthesis is not injective, so those inverse formulas cannot simply be applied. Section 4 supplies the exact quotient version needed here. The chapter's example $G_t = \operatorname{diag}(1, t^2)$, $\widehat{G}_t = \operatorname{diag}(1, t^2 + t)$ explains why small entry errors do not give relative control near a Gram kernel.

3.2 Replayable and residual-aware positivity

`C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/res-math/TR-item-047-astra.tex`

The sections "Algebraic and Moment Reformulation", "Ram Analysis and Synthesis Panels", "Global Optimality Sandwich", "Making the Moment and Identity Tests Exact", and "Residual-Aware SOS Lower Bounds" were read, including its worked nonconvex example. The theorem "Replayable global gap certificate", `eq:gap`, concerns a polynomial optimization gap, *not* a Hamiltonian spectral gap. Its applicable principle is a replayable exact identity plus independently certified positivity. The theorem "Inexact Gram and identity certificate", `eq:gramerror047`, deducts both an identity residual and a PSD defect. In our setting the corresponding deduction must multiply pair energy; a scalar error times the identity would not preserve zero modes. The weighted allowance and relative-error theorems below implement this distinction rather than merely renaming the source certificates.

3.3 Analysis, synthesis, and compatible coordinates

`C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`

Read passages are "Physical spaces precede matrices", "What a Ram representation means in this report" (`eq:moments`), "Ram moment coordinates that retain the complex" (`thm:transfer`), and "A concrete polynomial Ram construction". They specify $m = Ec$, an invertible synthesis $c = Rm$ or a left inverse only on compatible redundant data, and transport the physical mass as R^*MR. Here a finite physical space is $X_D = \operatorname{ran} W_D$, with measurements $\ell_r(x) = \langle w_{Dr}^*, x\rangle$, analysis W_D^*, and Gram data $G_D c$. These measurements are injective on X_D: if all vanish, x is orthogonal to its own span. Coefficients are determined only modulo $\ker W_D$. Selecting independent columns produces a valid synthesis on X_D, not an inverse on the original redundant copy space. The source's factorial diagonal congruence is a physical coordinate change in this sense. It does not change the best physical bound or create a new Laughlin state.

3.4 Finite-to-infinite safeguards

`C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01f-spectral-limits.tex`

The entire fragment was read, including `v21c01:pollution-proposition`, `v21c01:strong-not-norm`, `v21c01:stronger-contract`, and "Domain-aware approximate inversion". Its graph-core pollution example warns that exact finite matrices and strong convergence need not give spectral exclusion. Our source avoids that particular error by invariant fixed-degree blocks plus a uniform

coefficient, but still needs its full-Fock relative estimate. Finite Gram positivity alone supplies neither that estimate nor continuum completeness.

All these are private representations or verification disciplines whose mathematical ingredients include classical linear algebra, Galerkin metric transport, and SOS residual accounting. CAR, Schur averaging, Bessel's inequality, the spectral theorem, and rational PSD tests are classical. No historical novelty is claimed for them or for the Ram coordinate transport itself.

4 Exact certificates on a singular physical quotient

Lemma 4.1 (Compression without an inverse)**.** *Let $W : E \to X$ be linear between finite-dimensional complex Hilbert spaces, $G = W^*W$, and $C = C^*$. Then*

$$GCG \succeq 0 \quad \Longleftrightarrow \quad WCW^* \succeq 0.$$

Neither assertion is equivalent in general to $C \succeq 0$ on all E.

Proof. $\ker G = \ker W$ follows from $\langle x, Gx\rangle = \|Wx\|^2$. Taking orthogonal complements gives $\operatorname{ran} G = \operatorname{ran} W^*$. The two quadratic forms test $\langle y, Cy\rangle$ for exactly the same set of vectors y in that range. For the final warning take $W = (1\ 0)$ and $C = \operatorname{diag}(1, -1)$. □

This proves the source compression lemma afresh, but it is not itself a new improvement over that source. The useful extension is an exact rational reduced certificate with an independently checkable range witness.

Theorem 4.2 (Rational range factorization)**.** *Let $Z = Z^T \in \mathbb{Q}^{d\times d}$ and $B = B^T \in \mathbb{Q}^{d\times d}$. Let J consist of $k = \operatorname{rank} Z$ independent columns of Z and choose k row indices I for which J_I is invertible. Put*

$$R = J_I^{-1} Z_I, \qquad T = J^T B J.$$

Then $Z = JR$, $\operatorname{rank} R = k$, and

$$ZBZ = R^T T R, \qquad ZBZ \succeq 0 \iff T \succeq 0. \tag{13}$$

For $k = 0$, $Z = 0$ and the test is vacuous. To certify that k is the rank it suffices to supply a nonzero k-minor of J and the exact identity $Z = JR$.

Proof. Every column of Z is in $\operatorname{ran} J$, so restriction to rows I determines its unique coordinates in J. This proves $Z = JR$. Since Z has rank k, R has full row rank. Transposing $Z = JR$ and using symmetry gives $ZBZ = R^T J^T B J R$. One implication in (13) is immediate. For the reverse, every $y \in \mathbb{R}^k$ equals Rx for some x, so $y^T T y = x^T ZBZ x \geq 0$. Real symmetric positivity is equivalent to complex Hermitian positivity. The two rank witnesses give the lower and upper rank bounds respectively. □

In the source appendix put $u_r = 1/[r!(D-r)!]$, $D_u = \operatorname{diag}(u_r)$, and $S = \operatorname{diag}(\sqrt{u_r})$. The rational matrices Y, Z satisfy $E_D = SYS$, $G_D = SZS$. Therefore

$$G_D(2\chi - E_D + \varepsilon I)G_D = S(ZBZ)S, \quad B = \operatorname{diag}((2\mathbf{1}_{r=1} + \varepsilon)u_r) - D_u Y D_u. \tag{14}$$

Theorem 4.2 reduces this to $k_D \times k_D$ instead of $d_D^{\mathrm{copy}} \times d_D^{\mathrm{copy}}$, where $d_D^{\mathrm{copy}} = \lceil D/2 \rceil$ (not the spin dimension d_D). It requires no inverse of G_D, no numerical rank cutoff, and no radical arithmetic. The reduction is an algebraic simplification; no measured runtime improvement is asserted.

Proposition 4.3 (Characteristic-coefficient test)**.** *A real symmetric $k \times k$ matrix T is positive semidefinite if and only if every coefficient of* $\det(xI + T)$ *is nonnegative. These coefficients can be computed with $J_0 = I$ and*

$$X_b = TJ_{b-1}, \quad c_b = \frac{\operatorname{tr} X_b}{b}, \quad J_b = c_b I - X_b \quad (1 \le b \le k).$$

Proof. If the eigenvalues are nonnegative, their elementary symmetric sums are nonnegative. Conversely a negative real eigenvalue λ would produce the positive root $x = -\lambda$; a monic polynomial with nonnegative coefficients is strictly positive for $x > 0$. Symmetry is essential to the eigenvalue argument. The recurrence follows by writing $\operatorname{adj}(xI + T)$ as a matrix polynomial, comparing coefficients in $(xI + T)\operatorname{adj}(xI + T) = \det(xI + T)I$, and using $\frac{d}{dx}\det(xI + T) = \operatorname{tr}\operatorname{adj}(xI + T)$. These identities also apply to singular T as polynomial identities. □

Floating eigenvalue signs near zero are not used. An exact nullspace and a positive reduced test separate redundancy from instability. If T has additional zero directions, the same coefficient test remains valid without asserting strict positivity.

5 An explicit relative-error budget

5.1 Why absolute error is insufficient

Proposition 5.1 (Kernel compatibility)**.** *For $h \succeq 0$ and $E = E^*$ on a finite-dimensional Hilbert space, there exists $\rho < \infty$ with $-\rho h \preceq E \preceq \rho h$ if and only if $E \ker h = 0$. The least such ρ is $\|h^{\dagger/2} E h^{\dagger/2}\|$, where the inverse square root is taken on* $\operatorname{ran} h$ *and zero on* $\ker h$*.*

Proof. If the two form inequalities hold and $v \in \ker h$, then $\langle v, Ev\rangle = 0$. Testing $v + tu$ for real and purely imaginary t forces $\langle u, Ev\rangle = 0$ for every u, so $Ev = 0$. Conversely this condition makes E block diagonal with zero kernel block. Congruence by $h^{-1/2}$ on $\operatorname{ran} h$ converts the desired inequality into $-\rho I \preceq h^{-1/2} E h^{-1/2} \preceq \rho I$. The spectral theorem gives the asserted least constant. □

For example $W_t = \operatorname{diag}(1, t)$, $C = \operatorname{diag}(1, -1)$ gives $W_0 C W_0^* \succeq 0$ but $W_t C W_t^* = \operatorname{diag}(1, -t^2)$. Despite convergence in norm, it is not bounded below by $-\rho \operatorname{diag}(1, 0)$ for any finite ρ when $t > 0$. The source's pair-generated annihilators are what prevent this kernel incompatibility; continuity of Gram ranges would not.

5.2 A quantitative version of the source transfer

On $\mathcal{F}_{\mathrm{loc}}$ define, for $Q \ge 24$,

$$d_Q(x) = \sqrt{(Q)_x / Q^x}, \quad \Delta_Q |A\rangle = \prod_{x \in A} d_Q(x) |A\rangle, \quad r_p(Q) = \sqrt{(2Q)^p / (2Q - 2)_p}.$$

The exact source identities, obtained by tracking removed orbital labels in a basis wedge, are

$$B_p = r_p \Delta_Q^{-1} B(v_p^*) \Delta_Q, \tag{15}$$

$$B(v_p \wedge e_j \wedge e_k) = \frac{r_p}{d_Q(j) d_Q(k)} \Delta_Q^{-1} B(v_p^* \wedge e_j \wedge e_k) \Delta_Q. \tag{16}$$

All factors are positive, $r_p \geq 1$ and $0 < \Delta_Q \preceq I$. The inverse diagonal acts on the output particle sector, not the input. In particular

$$\Delta_Q h_* \Delta_Q \preceq h_Q, \qquad h_* = \sum_{p=0}^{23} B(v_p^*)^* B(v_p^*). \tag{17}$$

Indeed $B(v_p^*)\Delta_Q = r_p^{-1}\Delta_Q B_p$, and its norm on every vector is at most that of B_p.

Fix $D \in \{1, \ldots, 23\}$ and index $b = (p, j, k)$ by $p + j + k = D$, $p \geq 0$, $0 \leq j < k \leq 24$. Put

$$A_b = c_k c_j B(v_p^*), \quad C_D^Q = (d_D/d_a)\chi - E_D^Q + \varepsilon I, \quad \beta_{rb}^Q = S_r^Q(D, D; p, j, k)\frac{r_p}{d_Q(j)d_Q(k)}.$$

Let $M_Q = (\beta^Q)^T C_D^Q \beta^Q$ and $M_* = (\beta^*)^T(2\chi - E_D + \varepsilon I)\beta^*$, where $\beta_{rb}^* = S_r(D, D; p, j, k)$. These are real symmetric matrices on the uncoupled triple indices; M_* need not itself be PSD. For a symmetric matrix A write $\|A\|_{\mathrm{row}} = \max_b \sum_c |A_{bc}|$, so $\|A\|_2 \leq \|A\|_{\mathrm{row}}$.

Theorem 5.2 (Computable transfer budget)**.** *Suppose* (10) *holds for this D. Set*

$$\tau_Q = \prod_{x=0}^{24} \frac{Q^x}{(Q)_x} - 1, \tag{18}$$

$$a_D = \|M_*\|_{\mathrm{row}}, \qquad b_D(Q) = \|M_Q - M_*\|_{\mathrm{row}}, \qquad m_D = \lceil D/2 \rceil, \tag{19}$$

$$\rho_D(Q) = m_D((1 + \tau_Q)b_D(Q) + \tau_Q a_D). \tag{20}$$

Then on the entire $\mathcal{F}_Q$,

$$\sum_{r,s} (C_D^Q)_{rs} B(w_{Dr}^Q)^* B(w_{Ds}^Q) \succeq -\rho_D(Q) h_Q. \tag{21}$$

In particular $\rho_D(Q) \to 0$. The formulas involve finite coefficient matrices and a scalar product, not enumeration of the 2^{25} local occupation states. Moreover

$$0 \leq \tau_Q \leq \exp\left(\frac{2300}{Q - 23}\right) - 1. \tag{22}$$

Proof. Each eigenvalue of Δ_Q^{-2} is a subproduct of numbers $d_Q(x)^{-2} \geq 1$. Its largest is the full product, so its norm is $1 + \tau_Q$ and $\|\Delta_Q^{-2} - I\| = \tau_Q$. By (16) the left side of (21) is

$$\Delta_Q \sum_{b,c} (M_Q)_{bc} A_b^* \Delta_Q^{-2} A_c \Delta_Q.$$

The reference expression $\sum_{b,c}(M_*)_{bc} A_b^* A_c$ is positive: Lemma 4.1 and the assumed certificate give positivity on four particles, and the lift $\mathcal{L}_4$ preserves it. On the Hilbert direct sum indexed by b, the remaining block operator is

$$(M_Q - M_*) \otimes \Delta_Q^{-2} + M_* \otimes (\Delta_Q^{-2} - I),$$

whose norm is at most $(1 + \tau_Q)b_D(Q) + \tau_Q a_D$. For fixed p, at most $\lceil (D - p)/2 \rceil \leq m_D$ pairs $j < k$ satisfy $j + k = D - p$. Since $\|c_k c_j\| \leq 1$, $\sum_b \|A_b \phi\|^2 \leq m_D \langle \phi, h_* \phi \rangle$. Apply this with $\phi = \Delta_Q \psi$ and use (17). Spectator factorization proves the same inequality on $\mathcal{F}_Q$.

Finally,

$$\log(1 + \tau_Q) = \sum_{x=0}^{24} \sum_{a=0}^{x-1} -\log(1 - a/Q) \leq \frac{1}{Q - 23} \sum_{x=0}^{24} \frac{x(x-1)}{2} = \frac{2300}{Q - 23}.$$

The inequality uses $-\log(1-t) \leq t/(1-t)$ for $0 \leq t < 1$. Fixed-deficit convergence gives $b_D(Q) \to 0$, completing the proof. □

This is an explicit majorant replacing the source's unspecified norm of a block operator. The rational τ_Q can be evaluated exactly; finite-flux S^Q, E^Q, β^Q are algebraic numbers for integer Q, so outward rational enclosures of square roots give certified bounds for $b_D(Q)$. We have not implemented those enclosures or evaluated a flux threshold. An asymptotic $O(Q^{-1})$ coefficient requires proved rates for those fixed-deficit coefficients, not just the diagonal rate (22). The exponential bound can be extremely conservative at moderate flux and is not an efficiency claim.

If all retained three-body finite-flux tests and all four-body limiting tests hold, the source argument with these constants gives the explicit conditional inequality

$$H_Q^2 \succeq \left(1 - \eta - \delta_Q - 1222\varepsilon - 24 \sum_{D=1}^{23} \rho_D(Q)\right) H_Q. \tag{23}$$

There is no hidden particle-number factor. A claimed concrete threshold must certify the bracket and the finite-flux three-body inequalities, including their common lower cutoff in Q; testing one isolated Q does not prove a threshold for all larger fluxes.

6 Weighted allowances with exact diagonal witnesses

The source uses a scalar εI in every copy space. The next proposition permits nonuniform positive allowances while keeping their full-Fock cost explicit. It is a conditional certificate design tool, not an assertion that a smaller optimized cost has been found.

Proposition 6.1 (Anisotropic allowance)**.** *At a fixed Q, D let S^Q be the real coefficient matrix, in the source's chosen real coupling bases, of the orthonormal highest-copy vectors in the uncoupled triple basis, so $S^Q(S^Q)^* = I$. Let $\Gamma_D \succeq 0$ be a Hermitian copy matrix. If nonnegative scalars x_b satisfy*

$$\operatorname{diag}(x_b) \succeq (S^Q)^* \Gamma_D S^Q, \tag{24}$$

then

$$d_a \int \sum_{r,s} (\Gamma_D)_{rs} B(w_{Dr}^Q(g))^* B(w_{Ds}^Q(g))\, dg \preceq \left(\sum_b x_b\right) H_Q. \tag{25}$$

For $\Gamma_D = \varepsilon I$, $x_b = \varepsilon$ is feasible and recovers $n_D\varepsilon$. More generally $x_b = \sum_c |A_{bc}|$, with $A = (S^Q)^* \Gamma_D S^Q$, is a sufficient explicit choice.

Proof. Write $a_b = c_k c_j B_p \psi$. Because S^Q is real, conjugate-linearity of B gives the annihilator vector $S^Q a$ with these same coefficients. For a general complex coupling matrix one must instead use its entrywise conjugate in that identity and adjust the witness accordingly. The left integrand before averaging is the quadratic form of $A \otimes I$ on the Hilbert-space-valued vector $(a_b)_b$. The witness bounds it by $\sum_b x_b \|a_b\|^2 \le \sum_b x_b \|B_p\psi\|^2$. For each normalized pair v_p, Schur averaging gives $d_a \int B_p(g)^* B_p(g)\, dg = H_Q$. This proves (25). Since $S^*S \preceq I$, the scalar choice follows. For the row-sum choice, the diagonal entries of a PSD A are nonnegative, and $\operatorname{diag}(x) - A$ has diagonal $\sum_{c \ne b} |A_{bc}|$. Pairing off-diagonal terms and using $2|\overline{y}_b y_c| \le |y_b|^2 + |y_c|^2$ proves its positivity. □

An approximate diagonal witness is useful only if its own residual is included. For example a verified $\operatorname{diag}(x) - A \succeq -\zeta I$ is repaired by replacing each x_b by $x_b + \zeta$, costing $n_D\zeta$ after averaging. This is the operator counterpart of the private residual-aware SOS deduction, with the correct physical energy rather than a state count. Changing allowances also changes the transfer matrices C_D^Q and M_Q, so an optimized limit certificate must be revalidated in that transfer. Positivity only on a numerically selected range is insufficient.

6.1 A proved finite-flux support refinement

The independent audit supplied the following local strengthening. Its argument is included in full because a zero limiting Gram eigenvalue alone would not justify deleting a finite-flux block.

Proposition 6.2 (Support-aware allowance)**.** *For $Q \geq 24$, the scalar allowance in the retained four-body comparison costs at most $1123\varepsilon H_Q$ instead of $1222\varepsilon H_Q$. Subject to the remaining source comparisons, the limiting gap budget becomes*

$$\widetilde{\gamma}_* = 1 - \eta - \frac{61}{4096} - 1123\varepsilon = \frac{4646433319001}{10^{14}}. \tag{26}$$

This is not an endpoint estimate on a finite sphere.

Proof. For $p+j+k = D \leq 23$, $j < k$, the annihilator $c_k c_j B_p$ vanishes exactly when every pair $x < y$ with $x + y = p + 1$ intersects $\{j, k\}$. Indeed the nonzero coefficients (4) all have the same nonzero support for $Q \geq 24$; distinct surviving pairs produce distinct Slater quadruples in $v_p \wedge e_j \wedge e_k$ and cannot cancel. Removing zero coordinates from the Hilbert-space-valued vector in Bessel's inequality leaves its norm unchanged.

For $p = 0$ the only pair is $\{0, 1\}$, and among triples with $1 \leq p + j + k \leq 23$ there are 44 choices meeting that pair. For $p = 1$ the only pair is $\{0, 2\}$ and there are 41. For $p = 2$, the disjoint pairs $\{0, 3\}, \{1, 2\}$ have four two-element transversals; for $p = 3$ the pairs $\{0, 4\}, \{1, 3\}$ also have four. All fit the cutoff. For $p \geq 4$ there are at least three disjoint pairs, so two labels cannot intersect them all. Thus 93 of the 1222 triples vanish identically, leaving 1129.

In addition, the entire $D = 6$ highest-weight image vanishes at finite flux. Its orbital sum is 7, whose only increasing quadruple is $(0, 1, 2, 4)$. The raising operator sends that wedge to

$$\sqrt{4(Q-3)}\, e_0 \wedge e_1 \wedge e_2 \wedge e_3 \neq 0;$$

all other terms repeat an occupied index. Hence that weight space has no nonzero highest vector. Equivariance of the wedge map forces every w_{6r}^Q to vanish, and lowering forces its entire spin image to vanish. Removing its six remaining triples leaves $1129 - 6 = 1123$. The $D \leq 4$ images vanish already by the minimum quadruple sum 6. Rotational averaging of each surviving pair norm costs exactly H_Q, proving the allowance estimate. Combining with the unchanged two-body cost and tail gives (26). □

The same support information sharpens the explicit error account. Let $\mathcal{T}_D^+$ consist only of the nonzero triple annihilators, $\nu_{D,p} = \#\{(p, j, k) \in \mathcal{T}_D^+\}$, and $\nu_D = \sum_p \nu_{D,p}$. Restrict M_Q, M_* to these indices and put $e_D(Q) = (1 + \tau_Q) b_D(Q) + \tau_Q a_D$ for the resulting row norms. Keeping multiplicities in the proof of Theorem 5.2 gives

$$\sum_{r,s} (C_D^Q)_{rs} B(w_{Dr}^Q)^* B(w_{Ds}^Q) \succeq -e_D(Q) \sum_p \nu_{D,p} B_p^* B_p.$$

Consequently (23) improves, under the same remaining premises, to

$$H_Q^2 \succeq \left(1 - \eta - \delta_Q - 1123\varepsilon - \sum_{\substack{1 \leq D \leq 23 \\ D \neq 6}} \nu_D e_D(Q) \right) H_Q. \tag{27}$$

No finite-flux coefficient enclosures were computed, so this is a proved certificate formula, not a numerical threshold. The modest increase $297/10^6$ in the limiting coefficient is not treated as a standalone substantial advance.

7 An exact spherical particle-addition refinement

This independent local result sharpens source `cor:charge-neutral`. It needs the elementary unique zero mode at Laughlin flux, but does not assume a uniform spectral gap or any four-body positivity certificate.

Theorem 7.1 (Addition overlap of the spherical singlet)**.** *Let $N \geq 2$, $Q = 3(N-1)$, $m = Q+1 = 3N-2$, and let ψ_N be the normalized Laughlin vector in $\bigwedge^N U_Q$. Let P_N project onto its line, and define on $\bigwedge^{N+1} U_Q$*

$$A_N = \sum_{j=0}^{Q} c_j^* P_N c_j, \qquad \mathcal{A}_N = \operatorname{span}\{c_j^* \psi_N : 0 \leq j \leq Q\}.$$

Then $\dim \mathcal{A}_N = m$ and

$$A_N = \left(1 - \frac{N}{m}\right) P_{\mathcal{A}_N}. \tag{28}$$

If g_N is the first positive eigenvalue of $H_{N,Q}$, then

$$H_{N+1,Q} \succeq \frac{g_N}{N-1}\left[(N+1)I - \left(1 - \frac{N}{m}\right) P_{\mathcal{A}_N}\right]. \tag{29}$$

Consequently the fixed-flux charge gap satisfies

$$\Delta_N = E_0(N+1;Q) \geq \frac{N}{N-1}\left(1 + \frac{1}{3N-2}\right) g_N. \tag{30}$$

On vectors orthogonal to $\mathcal{A}_N$, the stronger form bound $(N+1)g_N/(N-1)$ holds. No invariance of that orthogonal complement under $H_{N+1,Q}$ is asserted.

Proof. Every bracket $u_i v_j - u_j v_i$ is $SU(2)$ invariant, so the normalized product ψ_N is a singlet. Its one-particle density matrix $\rho_{ij} = \langle \psi_N, c_j^* c_i \psi_N \rangle$ commutes with the irreducible one-particle representation. Schur's lemma and $\operatorname{tr}\rho = \langle \psi_N, \sum_i c_i^* c_i \psi_N \rangle = N$ give $\rho = (N/m)I$. The creation vectors therefore have Gram matrix

$$\langle c_i^* \psi_N, c_j^* \psi_N \rangle = \delta_{ij} - \langle \psi_N, c_j^* c_i \psi_N \rangle = \left(1 - \frac{N}{m}\right)\delta_{ij}.$$

Since $m > N$, these are m nonzero orthogonal vectors of the same norm. Summing their rank-one operators proves (28).

The pure two-body normal-ordered identity on $N+1$ particles is

$$\sum_j c_j^* H_{N,Q} c_j = (N-1) H_{N+1,Q}. \tag{31}$$

For completeness, on an n-particle sector a normal-ordered monomial with two creators and two annihilators satisfies $\sum_j c_j^* O c_j = (n-2)O$: commute the number operator past the two creators, or apply the CAR twice and sum the two contraction terms. Linearity yields (31). This is the classical particle-number recursion used in [4].

The finite-dimensional spectral theorem and unique kernel give $H_{N,Q} \succeq g_N(I - P_N)$. Applying the positive map $X \mapsto \sum_j c_j^* X c_j$, using $\sum_j c_j^* c_j = (N+1)I$ on this sector, and inserting (28) proves (29). Its least bracket coefficient is $N + 1 - (1 - N/m) = N + N/m$, giving (30).

The N-particle ground energy is zero. Some $c_j\psi_N$ is nonzero since the sum of their squared norms is N; every B_p commutes with c_j, so this is an $(N-1)$-particle zero mode. Thus $E_0(N-1;Q)=0$ as well and the second difference defining Δ_N reduces to $E_0(N+1;Q)$. The bracket-divisibility proof excludes an $(N+1)$-particle zero mode because degree $3N$ exceeds Q. All statements hold for each finite $N \geq 2$. □

The source uses the general fermionic estimate $\|A_N\| \leq 1$. Here its exact value is $1-N/(3N-2)$, and the full projector identity gives additional information about which trial vectors receive the stronger lower bound. The coefficient alone is not the basis of a high contribution score: this is a local refinement, and its relative improvement tends to one as $N \to \infty$. Conditional on the source large-N theorem, one may substitute $g_N \geq 1/25$ in (30); that substitution is not independently certified here. At $N=2, Q=3$, $g_2=1$ and $H_{3,3}=(5/2)I$, so the improved coefficient is exact, while the source rank-only estimate yields 2.

8 Bounded verification and arithmetic evidence

The local checker is `_audit/M546-author-checks.py` and its machine-readable result is `_audit/M546-author-checks.json`. It uses Python's exact `fractions.Fraction`, integer factorials, binomial sums, and rational Gaussian elimination. The initial attempt to import SymPy failed because that package was absent; no package was installed. The subsequent successful check has a 120-second internal wall-clock budget and fixed deficit bounds. Its recorded runtime is operational evidence, not a comparative speed benchmark.

8.1 Input and three-body reconstruction

The seven integer rows are read from source `verification/rows.txt`. Their $(t, 10^7\lambda)$ pairs are

$$(0,7181518),\ (1,-112991),\ (1,-4247311),\ (2,3438746),$$

$$(5,-98645),\ (5,-3160276),\ (7,1441784).$$

Every printed entry $pj:a$ is parsed as separate single-digit p,j; duplicate labels and violations of the declared mode ranges are rejected. With $P=10^7$,

$$\alpha_{pj} = \frac{a_{pj}}{P}\sqrt{\frac{2^{p-t}t!(p+j-t)!}{p!j!}}.$$

The source's integer binomial transform is

$$U(c,z,T,p) = \sum_{h=\max(0,p+z-T)}^{\min(p,z)} (-1)^{z-h}\binom{z}{h}\binom{T-z}{p-h}c^{p-h},$$

zero outside $0 \leq p,z \leq T$. Let $X_{\rho,T} = \sum_{p+j=T} a_{pj}^{\rho} U(2,z,T,p)$. Direct radical cancellation in (7) gives

$$P^2 e_z = \sum_{\rho}\sum_{T=\max(z,t_\rho)}^{15} \frac{2^z\binom{T}{z}}{3^T 2^{t_\rho}\binom{T}{t_\rho}} X_{\rho,T}\left(X_{\rho,T} + 2\ell_\rho U(2,z,T,t_\rho)\right). \tag{32}$$

The executed rational margins $m_z = \frac{3}{2}(3z-1)(-1/2)^z - e_z$ give:

z	2	4	5	6	7	8	9	10	11	12	13	14	15
$\lfloor 10^6 m_z \rfloor$	1298313	211317	249	250	249	250	250	250	249	250	250	6470	250

Thus all thirteen strict limiting inequalities agree exactly with the source. Also the checker verifies

$$\sum \ell_\rho^2 = 93527408868499, \quad 1 - \eta - 61/4096 - 1222(3/10^6) = 4616733319001/10^{14}.$$

8.2 Four-body rational reconstruction and its cutoff

For odd $r \le D$ put

$$V_r(D,T;p,j,k) = U(1,r,j+k,j)U(1,D-r,T-r,p),$$

interpreted as zero for $r > j+k$. The checker reconstructs the source matrix for $1 \le D \le 9$ by

$$P^2 Y_{rs} = -\sum_\rho \sum_{p,j;q,l:\, T \ge D} a_{pj}^\rho a_{ql}^\rho V_r(D,T;p,j,k) V_s(D,T;q,l,i) \cdot 2^{1-2T+p+q-t_\rho+(r+s)/2} \frac{i!k!t_\rho!}{(T-D)!}, \tag{33}$$

where $i = p+j-t_\rho$, $k = q+l-t_\rho$, $T = p+j+k$; both entry sums are ordered. For each increasing quadruple A of nonnegative indices with sum $D+1$, put $f(A) = \prod_{a\in A} a!$ and

$$L_r(A) = 2^{(1-2D+r)/2} \sum_{\substack{x<y \text{ in } A \\ j<k,\ \{j,k\}=A\setminus\{x,y\}}} \sigma(x,y,j,k)(x-y) \cdot p!j!k! V_r(D,D;p,j,k), \quad p = x+y-1, \tag{34}$$

$$Z_{rs} = \sum_A \frac{L_r(A)L_s(A)}{f(A)}. \tag{35}$$

Here σ is the parity of sorting the four listed indices. All exponents are integers. Formula (35) is reconstructed for *every* $D \le 23$; its rank is certified by rational pivots and the exact range-factorization identity of Theorem 4.2. The resulting ranks are

D	1	2	3	4	5	6	7	8	9	10	11
k_D	0	0	0	0	1	0	1	1	2	1	3
D	12	13	14	15	16	17	18	19	20	21	22
k_D	2	4	3	5	4	6	5	7	6	8	7
D	23										
k_D	9										

For $D \le 9$ the checker also verifies exact symmetry of Y, forms $B, M = ZBZ, T = J^T B J$, verifies $M = R^T T R$, and checks all characteristic coefficients of both M and T are nonnegative. The source sign strings are reproduced as

`0, 0, 00, 00, +00, 000, +000, +000, ++000.`

These have dimensions $1,1,2,2,3,3,4,4,5$ respectively; the reduced nonzero tests have dimensions $1,1,1,2$. The JSON contains the exact rational reduced coefficients, not just their displayed signs. Ranks for $D \ge 10$ do *not* verify their positivity matrices; their Y and M were deliberately not replayed. There is no many-particle diagonalization, no stochastic test, and no use of floating eigenvalues to decide a zero sign.

8.3 Independent full finite-certificate replay

The separately authored sidecar `_audit/M546-independent-20261007/audit.txt` and checker `check_exact.py` were read in full. Its `exact-results.json` records a bounded 120-second-budget run that reconstructed all $D = 1, \dots, 23$ matrices Y, Z, B, M, reproduced all source characteristic sign strings, checked the zero terminal Cayley–Hamilton remainder, and established positive exact LDL pivots for every nonempty reduced range matrix. The full matrices, pivots, and coefficients are stored, not merely pass flags. Its limiting Gram ranks agree with the author's separate reconstruction above.

The sidecar additionally checked 624 squared pair-conjugation identities and 96 pair norms at $Q = 24, 25, 48, 100$, and the support counts used in Proposition 6.2. The surviving counts for $D = 1, \dots, 23$ are

$$0, 0, 0, 0, 6, 6, 11, 15, 21, 25, 32, 38, 45, 52, 60, 68, 77,$$
$$86, 96, 106, 117, 128, 140.$$

They sum to 1129, or 1123 after the analytically justified deletion of $D = 6$. The author also rechecked the supplied matrix products and reduced positivity with an acceptance checker and tested the $N = 2, Q = 3$ particle-addition identity exactly; its scope and result are recorded in `_audit/M546-author-review-checks.json`.

The two PSD methods in the sidecar share their assembled matrices: they are not independent constructions of the underlying physical operators. Its all-block replay certifies the printed finite arithmetic, not every spin-coupling derivation, the finite-sphere transfer thresholds, or the source's global theorem. The sidecar did not read the planar section or this author's report when producing that source audit. Parent review of the present local theorems is a separate finalization requirement.

9 Public comparison, limitations, and research directions

Primary public sources were consulted by public mathematical title or topic, with no private corpus text in a query. Ortiz et al. [2] develop the second-quantized pairing formulation and its zero-mode structure. The source's diagonal occupation-number change fits that classical geometry-dependent framework; the critical extra requirement for a gap is a uniform relative error bound. Nachtergaele–Warzel–Young [3] establish uniform gaps for a truncated pseudopotential. That result is not a certificate for the full spherical interaction, and neither our local mode set nor the source auxiliary squares license such an identification.

The public Lemm–Nachtergaele–Warzel–Young recursion [4], Theorem 2.1(I) and Lemma 2.3, controls charge versus neutral energies through many-body overlap matrices; it still requires model-specific information. Section 7 supplies the exact spherical singlet overlap before any ground-space compression. It does not claim a new general recursion. Rai et al. [5] develop a hierarchy of gap certificates for frustration-free spin systems. This situates the quadratic inequality method but does not verify the source's seven fermionic rows. Peyrl–Parrilo [6] develop exact rational SOS reconstruction and discuss strict-feasibility requirements. Here exact rank reduction addresses the prescribed singular Gram image; no generic rounding theorem is invoked across an unknown rank transition.

Advantages. Exact quotient witnesses distinguish redundancy from physical instability; the relative budget has the required zero-mode behavior; weighted allowances keep their cost independent of particle number; and the singlet projector yields an explicit local refinement that does not depend

on the unverified global theorem. The checks expose their entire finite scope and can fail by an exact negative numerator.

Costs and limits. Exact fraction sizes can grow rapidly. Rank reduction need not lower total runtime after matrix construction, and none was benchmarked. The bound on Δ_Q^{-2} uses the worst local occupation and may be loose. The finite-flux coefficient enclosures and any concrete all-large-Q threshold are unmet computational dependencies. Analytic reading of spin formulas is not a formal proof-assistant check. We establish no disorder stability, torus theorem, arbitrary-interaction extension, or historically new resolution of the Laughlin conjecture. The catalogue family title does not authorize importing a disorder result absent from this source.

Prioritized next work. First cross-validate the printed four-body arithmetic against an independent physical construction with separately derived phases. Next enclose the finite-flux coefficients and certify a tail and threshold uniformly on $Q \geq Q_0$, rather than reporting isolated large-Q samples. Then optimize diagonal allowance witnesses and include their reconstruction residuals in (23). Finally examine whether additional rotational information controls the compression of $P_{\mathcal{A}_N}$ to actual positive-energy ground spaces. None of those research tasks is credited as a realized result.

10 Verification status and contribution assessment

The realized contribution is the combination of the exact rational range certificate in Theorem 4.2, the explicitly computable pair-energy budget in Theorem 5.2, the weighted allowance witness in Proposition 6.1, and the exact singlet addition projector with its energy consequence in Theorem 7.1. The first three specialize classical certificate tools to the source's singular wedge maps and spectator extension. The fourth is a concrete source-level strengthening of the rank-only overlap estimate. This merits an internal C3 assessment of notable local contribution, not C4 or C5 based on report length, applicability, or the numerical coefficient alone. The independently proved support deletion and all-block rational replay strengthen the local evidence without resolving the remaining global dependencies.

The source's headline gap has not been independently verified. Only the stated bounded arithmetic was executed. Review readiness, parent findings and responses, native-editor/compiler responses, disk PDF export, visual inspection, and final freeze hashes are recorded in the local author audit sidecars. Final freezing requires parent mathematical review; a compiled draft alone is not a final delivery. This assessment is internal evidence-based assessment, not external peer review, and does not assert historical novelty.

A Reproducibility and read-scope ledger

The complete source read comprises `build/main.tex`; every file `build/sections/01-introduction.tex` through `build/sections/08-arithmetic.tex`; `build/figures/relative-transfer.tex`; `verification/rows.txt`; `build/references.bib`; and `README.md`. The appendix was read through its rational U, V, Y, L, Z, B, M formulas, recurrence, all sign strings, and final certificate remark. The source checkout supplied TeX and row data; no claim is made of a separate source-PDF visual reading. All mathematical source inputs are SHA-256 hashed in the checker result. The four private files and the exact ranges read are hashed and recorded separately in the author provenance record.

The bounded check command is

`C:/Users/mural/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/python.exe _audit/M546-author-checks.py`.

The standalone report has no external TeX inputs, bibliography build, or generated-matrix dependencies. Installed MiKTeX export uses the existing study helper `_tools/compile_report.py`; the author's owned build directory is `_build/TR-M546-Ram-Laughlin-Gap`. The requested maximum of two formatting QA rounds is separate from mathematical review. A final manifest binds report source, PDF, checker, executed results, relevant inputs, and review evidence, with no subsequent edits to frozen artifacts.

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